

The empirical recurrence for OEIS A233985: recovering a conjecture that is not written down

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Abstract

OEIS A233985 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 3125$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233985 is “Number of $(n+1) \times (4+1)$ 0..4 arrays with every 2×2 subblock having the sum of the squares of the edge differences equal to 10, and no two adjacent values equal.”. It has offset 1 and begins

4064, 62040, 914000, 14319916, 218943540, 3423714364, 52813238148, 823369058112, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:17:57, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 3125$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 3125$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 6250$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	11	c_{21}	277941067281430983839	c_{41}	10766487243264544883950574655
c_2	671	c_{22}	192021090433179403002	c_{42}	−12103130478389625471303011659
c_3	−7148	c_{23}	−5630746042984838229899	c_{43}	−37739901230894788768346574077
c_4	−196222	c_{24}	−1684731545963177627645	c_{44}	43922005667273826981314426022
c_5	2064010	c_{25}	94495261060346691987625	c_{45}	110575897971636541894547510603
c_6	33474648	c_{26}	−1538073525122638748110	c_{46}	−13149773513705863791554869791
c_7	−353835307	c_{27}	−1321659263658774726352650	c_{47}	−26941285905856575586932435816
c_8	−3764049203	c_{28}	360062163497652617120165	c_{48}	323567557547236304029277498219
c_9	40650612984	c_{29}	15477580232721466827042167	c_{49}	542264972647878576363628702334
c_{10}	298501346004	c_{30}	−7452263069321613499050976	c_{50}	−65041747619242003049509038410
c_{11}	−3350087102684	c_{31}	−152276496274567029057223478	c_{51}	−89423066913180840158113938044
c_{12}	−17409115161298	c_{32}	99357055459078777432406722	c_{52}	1059060717912042725438353331406
c_{13}	206917853426291	c_{33}	1261546592482552425448245426	c_{53}	1195863113307459087591872685036
c_{14}	766776328265960	c_{34}	−999372275052187921654233104	c_{54}	−138127682390191411084425294551
c_{15}	−9874772522970781	c_{35}	−8811841191757088483006935416	c_{55}	−128061949017034309355100599715
c_{16}	−25896367264643979	c_{36}	7981166748111020991943285224	c_{56}	1422261090384182237251509723238
c_{17}	372304172268874184	c_{37}	51909995498502203316980124350	c_{57}	1081201137537946257697294723799
c_{18}	673489207227761962	c_{38}	−51756766761841424199020379996	c_{58}	−113490408855813106230093638338
c_{19}	−11276597858044127256	c_{39}	−257728500038083903378469207588	c_{59}	−70598319176202249893667120969
c_{20}	−13335274368576418056	c_{40}	275539381769435755934679108948	c_{60}	685336427384310982377125076910

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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